Investments Primer

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Asset Pricing

- How to measure, price & hedge risk?

- Bonds
  - Risk-free loan
  - US Treasuries
- Stock
  - Ownership in a firm
  - Potentially receive future dividends and price appreciation
  - Risky
- Derivatives
  - Payoffs function of prices of other assets

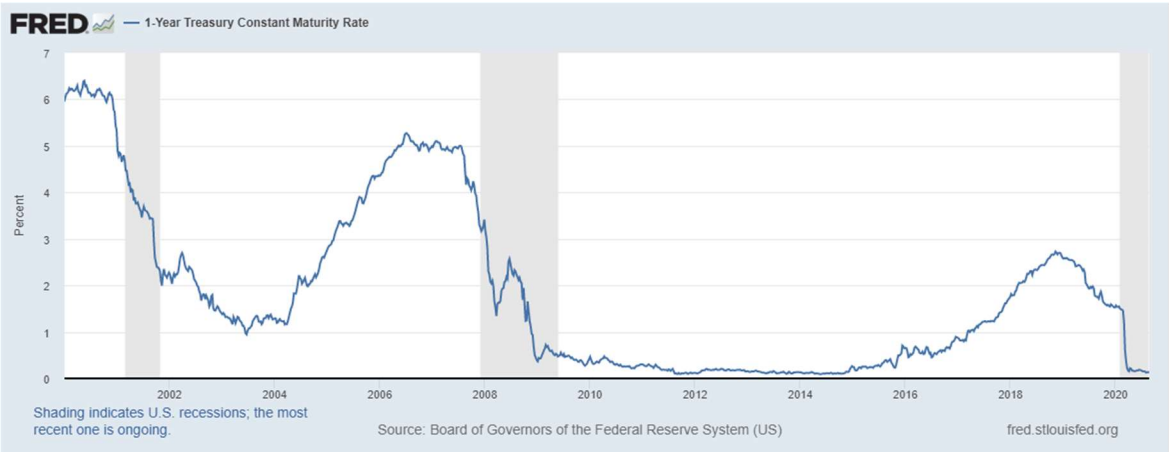
Investors' Motivation

- Consumption smoothing
  - Use assets to store wealth and transfer consumption to the future
- Risk allocation
  - Select assets consistent with their preferences for risk

Example: Consumption Smoothing

- Risk-free rate
  - \$100 today vs. \$100$+x$ in 1 year

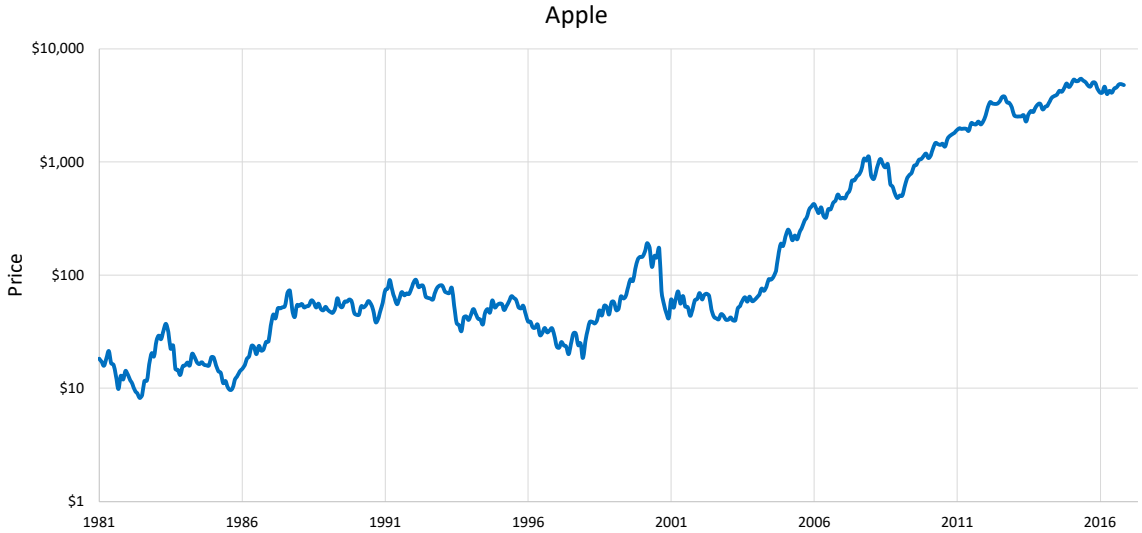
Current 1-Year US Treasury Rate



Example: Risk Allocation

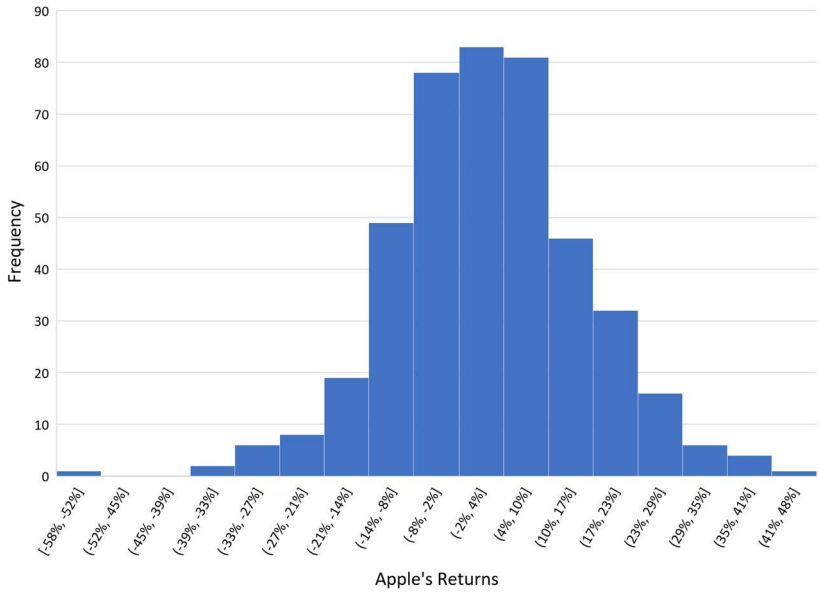
- Risk premium
  - \$100 with certainty vs.
  - Gamble:
    - Win \$12,000$+x$ with 50% probability
    - Lose \$10,000 with 50% probability

Apple's Price



Apple's Returns

$$R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$



Decompose Returns

$$R = R_f + RP$$

- Risk-free rate R_f
  - The risk-free rate reflects the time value of money
  - How willing an investor is to give up $1 today for $$$ in the future
  - The more impatient a consumer, the more future compensation he will demand to delay consumption and hence the higher the risk-free rate
- Risk premium RP
  - Investors who dislike risk are risk averse and demand risk premiums for bearing risk
  - The risk-premium increases when:
    - The riskiness of an asset increases
    - Investors become more risk averse

Central Theme & Struggle of Finance

How to allocate scarce resources? What to invest in?

- What is the appropriate price of an asset?
- Risk-reward trade-off
  - Investors demand higher expected returns for riskier securities
  - How do we measure, price and hedge risk?
- Risk is time-varying
  - How do we confront this as both economists and investors?

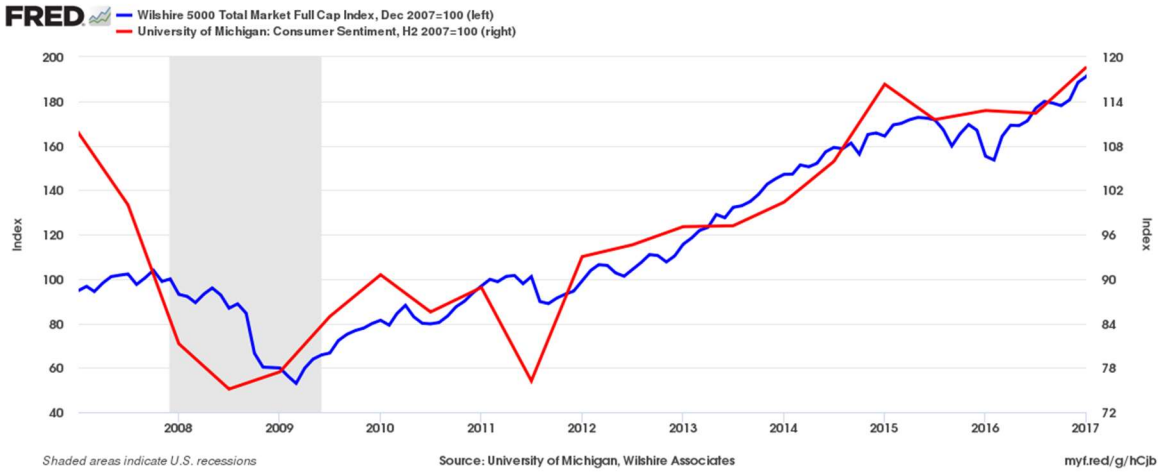
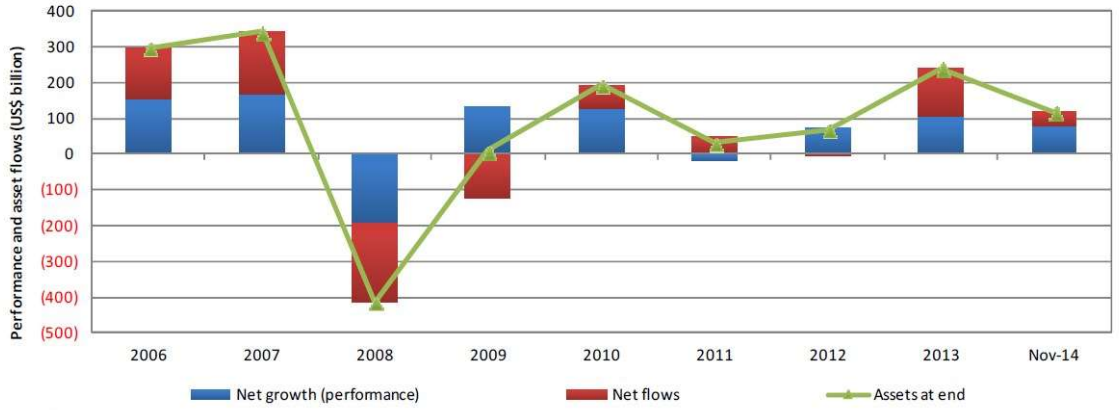


Figure 2b: Industry growth over the years



Source: Eurekahedge